

- Description

This module manages a reservoir water level monitoring system, ensuring that:

1. **The water level is continuously measured and compared against safe operational limits.**
2. **If the water level exceeds the maximum threshold, an overflow warning is triggered.**
3. **If the water level drops below the minimum threshold, a drought risk alert is issued.**
4. **A log of all water level readings, alerts, and corrective actions is maintained for water resource management.**

This system helps optimize water usage, prevent flooding, and ensure a stable water supply for irrigation and drinking water.

- Specification

module ReservoirWaterLevelMonitoring

var currentLevel, maxLevel, minLevel: real

var levelLog: sequence of string

process updateWaterLevel(changeAmount: real)

functional scenario form:

currentLevel <= maxLevel && levelLog = levelLog $\oplus$ ("Water Level Updated: " + currentLevel + "m"), Water level updated successfully.

||

currentLevel > maxLevel && levelLog = levelLog $\oplus$ ("Overflow Warning: " + currentLevel + "m, Immediate Water Release Required."), Overflow warning triggered, immediate action required.

||

currentLevel < minLevel && levelLog = levelLog $\oplus$ ("Drought Alert: " + currentLevel + "m, Water Conservation Recommended."), Drought risk alert triggered, water conservation recommended.

end process

end module

- Java Code

```
package code;
import java.util.*;
class Reservoir {
    private double maxLevel;
    private double minLevel;
    private double currentLevel;
    private List<String> levelLog;
    public Reservoir(double maxLevel, double minLevel, double initialLevel) {
        this.maxLevel = maxLevel;
```

```

        this.minLevel = minLevel;
        this.currentLevel = initialLevel;
        this.levelLog = new ArrayList<>();
    }
    public void updateWaterLevel(double changeAmount) {
        currentLevel += changeAmount;
        String message = String.format("Current Water Level: %.1f meter
s", currentLevel);
        if (currentLevel > maxLevel) {
            message += " | WARNING: Overflow Risk! Immediate Water Rele
ase Required.";
        } else if (currentLevel < minLevel) {
            message += " | ALERT: Drought Risk! Water Conservation Reco
mmended.";
        }
        System.out.println(message);
        levelLog.add(message);
    }
    public void printLevelLog() {
        System.out.println("\nWater Level Log:");
        for (String log : levelLog) {
            System.out.println(log);
        }
    }
}

public class c33_ReservoirSystem {
    public static void main(String[] args) {
        Reservoir reservoir = new Reservoir(50.0, 10.0, 30.0);
        // testcase-VT:
        //     reservoir.updateWaterLevel(79.2);
        //     reservoir.updateWaterLevel(-120.3);
        //     reservoir.updateWaterLevel(40.3);

        // testcase-FT:
        //     reservoir.updateWaterLevel(1.3394576298974116E307);
        //     reservoir.updateWaterLevel(-1.7976931348623157E308);
        //     reservoir.updateWaterLevel(4.482844445243999E290);
        // testcase-Z3:
        reservoir.updateWaterLevel(79);
        reservoir.updateWaterLevel(120);
        reservoir.updateWaterLevel(40);

        // testcase-UVT:

```

```
//      reservoir.updateWaterLevel(55);
//      reservoir.updateWaterLevel(-13);
//      reservoir.updateWaterLevel(30);

//      reservoir.printLevelLog();
}
}
```

- Coverage report

1. Test case number: 3
2. V-Method testing branch coverage: 100%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.7 %	6	838	844
▼ src	0.7 %	6	838	844
▼ code	0.7 %	6	838	844
▼ c33_ReservoirSystem.java	100.0 %	6	0	6
> c33_ReservoirSystem		0	0	0
> Reservoir	100.0 %	6	0	6

3. Vibration testing instruction coverage: 97.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	100	21,110	21,210
▼ src	0.5 %	100	21,110	21,210
▼ code	0.5 %	100	21,110	21,210
▼ c33_ReservoirSystem.java	97.1 %	100	3	103
> c33_ReservoirSystem	86.4 %	19	3	22
> Reservoir	100.0 %	81	0	81

4. Fuzz testing branch coverage: 83.3%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.6 %	5	837	842
▼ src	0.6 %	5	837	842
▼ code	0.6 %	5	837	842
▼ c33_ReservoirSystem.java	83.3 %	5	1	6
> Reservoir	83.3 %	5	1	6
> c33_ReservoirSystem		0	0	0

5. Fuzz testing instruction coverage: 97.1%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.5 %	100	21,110	21,210
▼ src	0.5 %	100	21,110	21,210
▼ code	0.5 %	100	21,110	21,210
▼ c33_ReservoirSystem.java	97.1 %	100	3	103
> c33_ReservoirSystem	86.4 %	19	3	22
> Reservoir	100.0 %	81	0	81

6. Z3 branch coverage: 16.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.1 %	1	839	840
▼ src	0.1 %	1	839	840
▼ code	0.1 %	1	839	840
▼ c33_ReservoirSystem.java	16.7 %	1	5	6
> Reservoir	16.7 %	1	5	6
> c33_ReservoirSystem		0	0	0

7. Z3 instruction coverage: 69.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	70	22,322	22,392
▼ src	0.3 %	70	22,322	22,392
▼ code	0.3 %	70	22,322	22,392
▼ c33_ReservoirSystem.java	69.3 %	70	31	101
> Reservoir	65.4 %	53	28	81
> c33_ReservoirSystem	85.0 %	17	3	20

8. Unimproved V-method branch coverage: 16.7%

Element	Coverage	Covered Branches	Missed Branches	Total Branches
▼ VibrationTestingTool	0.1 %	1	839	840
▼ src	0.1 %	1	839	840
▼ code	0.1 %	1	839	840
▼ c33_ReservoirSystem.java	16.7 %	1	5	6
> Reservoir	16.7 %	1	5	6
> c33_ReservoirSystem		0	0	0

9. Unimproved V-method instruction coverage: 69.3%

Element	Coverage	Covered Instructions	Missed Instructions	Total Instructions
▼ VibrationTestingTool	0.3 %	70	22,322	22,392
▼ src	0.3 %	70	22,322	22,392
▼ code	0.3 %	70	22,322	22,392
▼ c33_ReservoirSystem.java	69.3 %	70	31	101
> Reservoir	65.4 %	53	28	81
> c33_ReservoirSystem	85.0 %	17	3	20